Complete Orthonormal System (CONS) in Hilbert Space

An orthonormal set $\{e_i\}_{i \in I}$ is complete if:

$\langle x, e_i \rangle = 0$ for all $i \Rightarrow x = 0$

Parseval's Equality: $\|x\|^2 = \sum_{i \in I} |\langle x, e_i \rangle|^2$

Fourier expansion: $x = \sum_{i \in I} \langle x, e_i \rangle e_i$

Examples of CONS:

$\mathbb{R}^n$	*Standard basis:* $\{e_1, e_2, \ldots, e_n\}$
ℓ^2	$e_i = (0, \ldots, 1, \ldots, 0)$ *with 1 in position i*
$L^2[0, 2\pi]$	*Trigonometric basis:* $\{1, \cos(nx), \sin(nx)\}$